

Highlights Snapshot

National Taiwan University of Science and Technology
Master in Electrical Engineering

Taipei, Taiwan

Aug 2024 - Jun 2026 (Expected)

Paper

- *Syn3DTxt*, CVPRw 2025
- *MART*, IJCAI 2026 under review

Experience

TA for graduate courses Deep Learning and Computer Vision

Tech Stack

Python OpenCV Pandas PyTorch Git Hugging Face Latex TOEIC: 815



Syn3DTxt: Embedding 3D Cues for Scene Text Generation

Li-Syun Hsiung¹, Jun-Kai Tu¹, Kuan-Wu Chu², Yu-Hsuan Chiu¹, Yan-Tsung Peng², Sheng-Luen Chung¹, Gee-Sern Jison Hsu¹

¹National Taiwan University of Science and Technology ²National Chengchi University



MOTIVATION

- Challenge in Scene Text Editing/ Generation: State-of-the-art models rely on 2D planar datasets → fail under perspective & curved text. (Fig. 1.)
- Goal: perspective-aware, high-fidelity scene-text generation for editing or rendering tasks.



Fig. 1. Example of previous Dataset

CONTRIBUTIONS

- Syn3DTxt: 150 k synthetic images with RGB-encoded normal maps, arc bends, and single/dual/triple-axis rotations.
- A public 3Daware pipeline for synthetic data generation with per-pixel surface-normal cues, designed for future 3D text datasets. (Fig. 2.)
- 15 % ↑ SSIM & 17.7 % ↓ FID versus 2D baselines on perspective-consistent text editing.

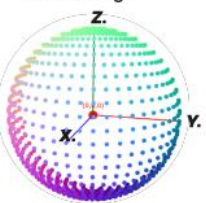


Fig. 2. Visualization of RGB-encoded surface normals on a sphere.

OUR METHOD

- 3D Rotation Pipeline: apply roll (γ) → pitch (θ) → yaw (ϕ) to a flat text mask, respecting human viewpoint order. (Fig. 3.)
- Arc Distortion: simulate curved surfaces by bending text at $0^\circ, 60^\circ, 120^\circ$. (Fig. 4.)
- Surface-Normal Encoding: map $(nx, ny, nz) \rightarrow (R, G, B) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$.
- Dataset Stats: 70 fonts, 150 k train / 2 k eval / 2 k advanced / 2 k curved.

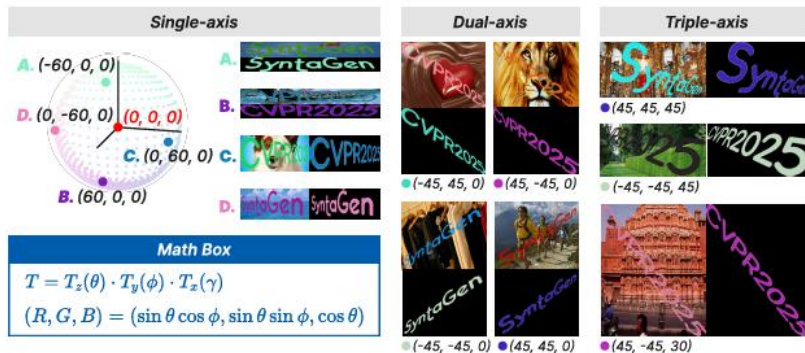


Fig. 3. RGB-encoded normal vectors within a sphere system. Text images rendered at matching orientations, grouped by single-axis, dual-axis, and triple-axis rotations.



Fig. 4. 3D bending synthesis.

Top: rendered text | Middle: RGB normal mask | Bottom: binary text mask.
Distinct normals assigned to each character capture curved-surface geometry far better than planar rotation baselines.

EVALUATION

- Baseline: MOSTEL-150 k (2D mask) → fine-tuned with Syn3DTxt-150 k (3D mask).
- Test Sets: Syn3DTxt-eval-2 k, -advanced (large rotations), -wrap (curved), Tamper-Syn2k.
- Metrics: SSIM ↑ +15 %, PSNR ↑, MSE ↓, FID ↓ 17.7 % vs. 2D fine-tune.
- Qualitative: 3D model preserves stroke thickness under yaw/pitch (Fig. 5); Comparison between the original MOSTEL and our enhanced MOSTEL 3D on ScenePair across three representative case. (Fig. 6.)
- Outperforms SRNet (GAN) & TextCtrl (diffusion) on perspective benchmarks. (Table 1.)

Models	Syn3DTxt-eval-2k				Syn3DTxt-wrap				Syn3DTxt-eval-advanced				Tamper-Syn2k			
	PSNR ↑	SSIM ↑	MSE ↓	FID ↓	PSNR ↑	SSIM ↑	MSE ↓	FID ↓	PSNR ↑	SSIM ↑	MSE ↓	FID ↓	PSNR ↑	SSIM ↑	MSE ↓	FID ↓
SRNet [14]	17.011	0.5283	0.0234	80.502	16.433	0.5027	0.0267	61.832	17.152	0.5259	0.0228	87.333	18.042	0.6114	0.0216	51.538
TextCtrl [16]	17.837	0.6067	0.0293	36.288	16.646	0.5371	0.0266	40.990	18.523	0.6302	0.0188	34.800				
MOSTEL [11]	20.527	0.7265	0.0119	40.005	17.386	0.6185	0.0179	45.830	19.855	0.7677	0.0133	41.311	20.489	0.7912	0.0128	36.337
MOSTEL + 2D Finetuned	20.846	0.7315	0.0103	33.991	17.196	0.6090	0.0188	37.625	21.256	0.7651	0.0114	34.803	19.746	0.6666	0.0157	41.921
MOSTEL + 3D Finetuned	21.358	0.8151	0.0093	29.834	18.552	0.7251	0.0175	34.086	22.133	0.8326	0.0083	29.174	19.790	0.7663	0.0156	40.420
MOSTEL 3D from scratch	21.256	0.7630	0.0097	28.790	18.592	0.6266	0.0173	35.000	21.976	0.7801	0.0084	28.639	19.698	0.6661	0.0157	42.897

Table 1. Quantitative results on our dataset and Tamper-Syn2k.



Fig. 5. With vs. Without 3D normals: models trained using 3D normal information render sharper, perspective-correct text than the 2D baseline.

CONCLUSION

- Syn3DTxt injects 3D surface-normal cues into synthetic data, delivering markedly more realistic scene-text generation.
- Models trained on Syn3DTxt handle perspective distortion and non-planar text with higher fidelity, advancing robustness and generalization in scene-text research.



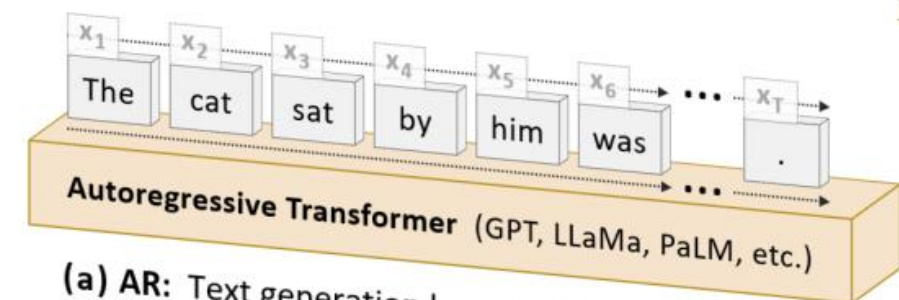
MART: Multi-Modal AutoRegressive Transformer for High-Fidelity Scene Text Synthesis

Li-Syun Hsiung, Jun-Kai Tu, Gee-Sern Jison Hsu (IJCAI), 2026 **under review**

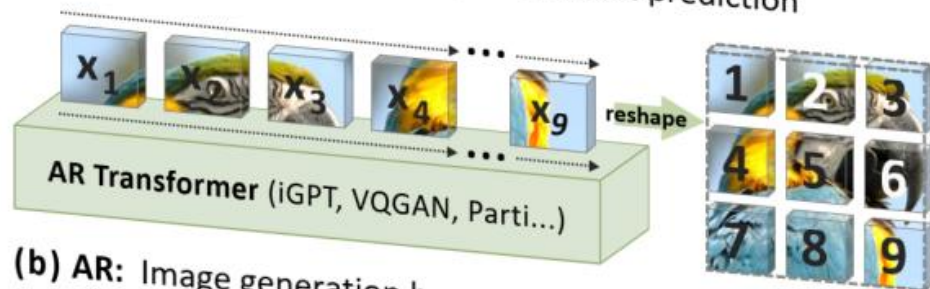


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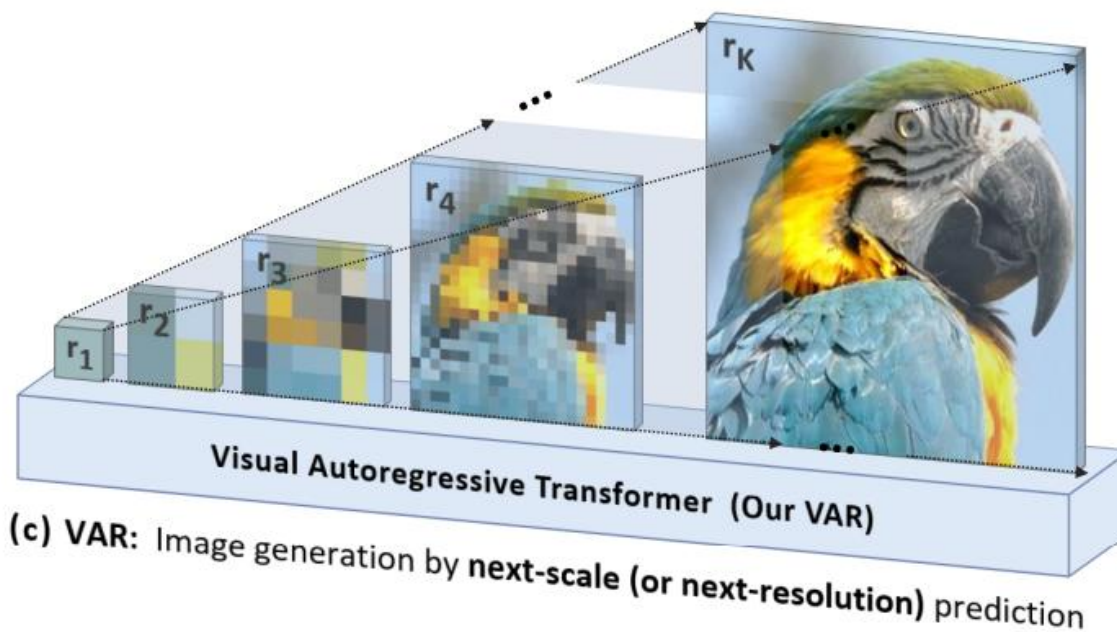


(a) AR: Text generation by **next-token** prediction



(b) AR: Image generation by **next-image-token** prediction

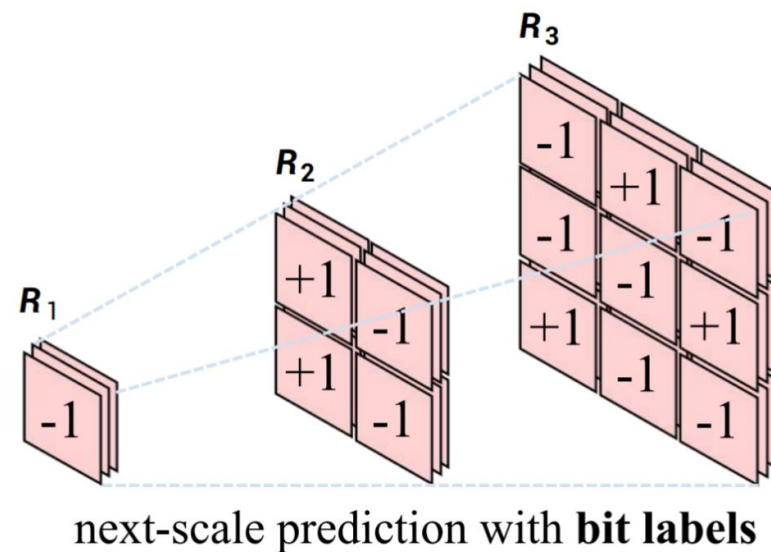
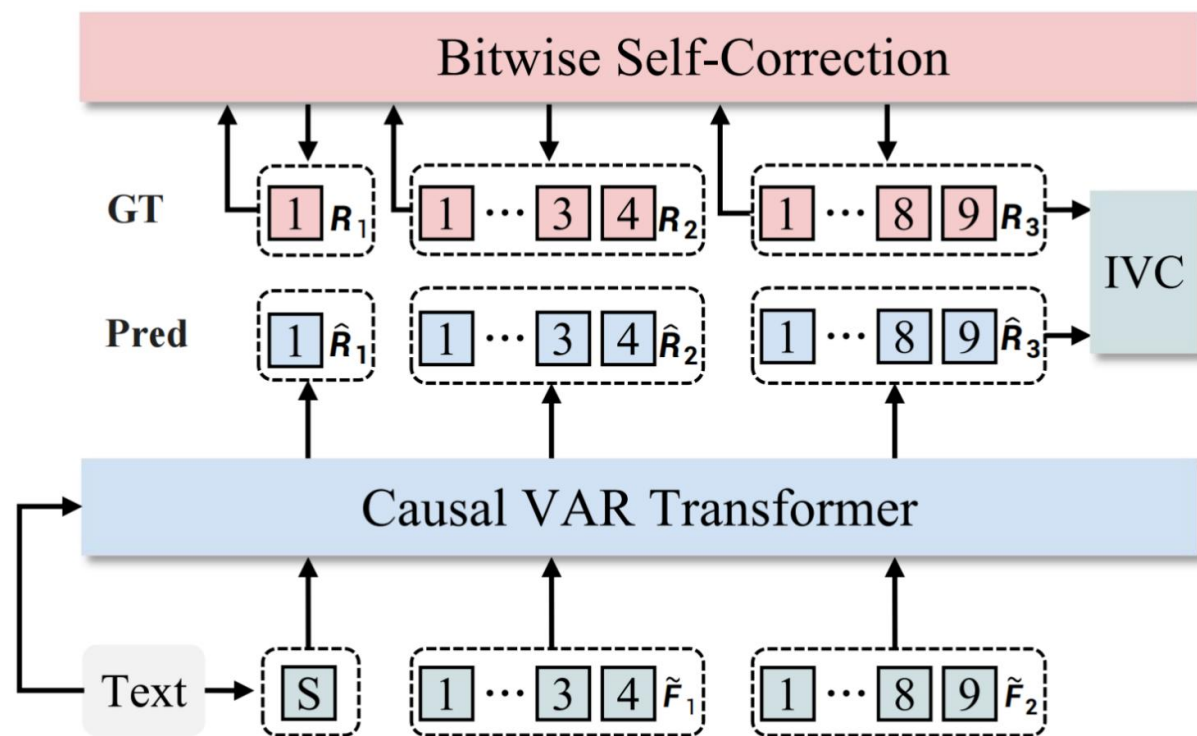
Three Different Autoregressive Generative Models



(c) VAR: Image generation by **next-scale (or next-resolution)** prediction

MART: Multi-Modal AutoRegressive Transformer for High-Fidelity Scene Text Synthesis

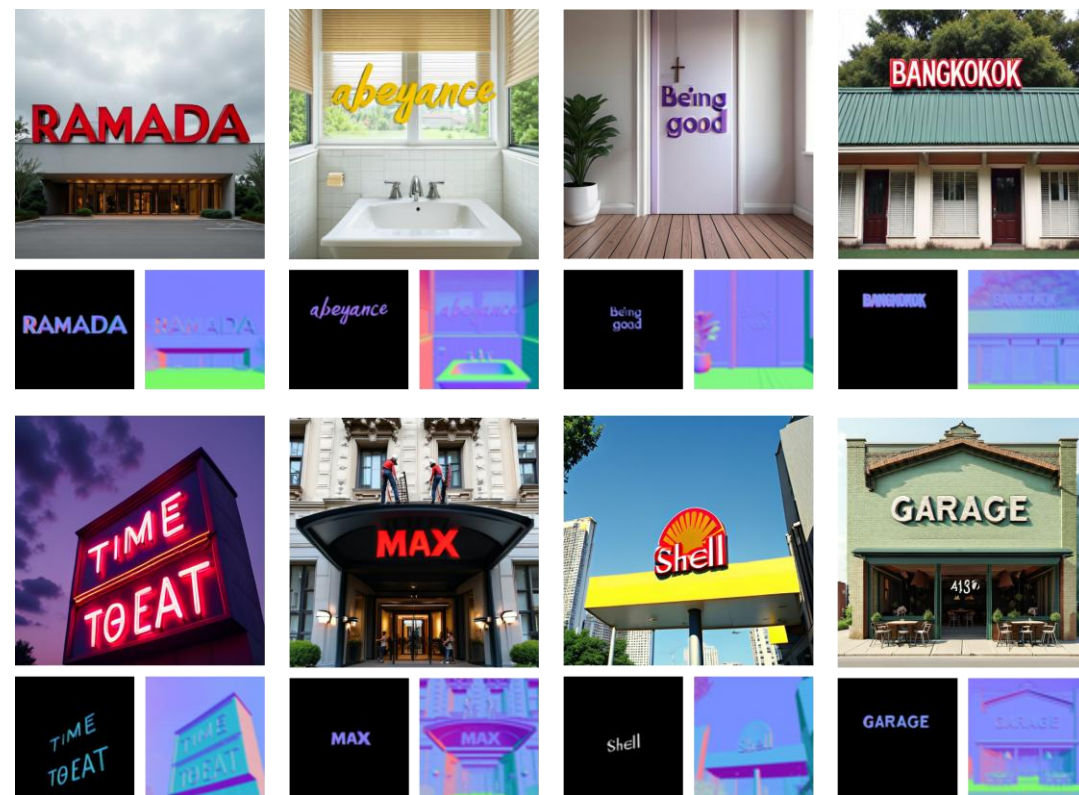
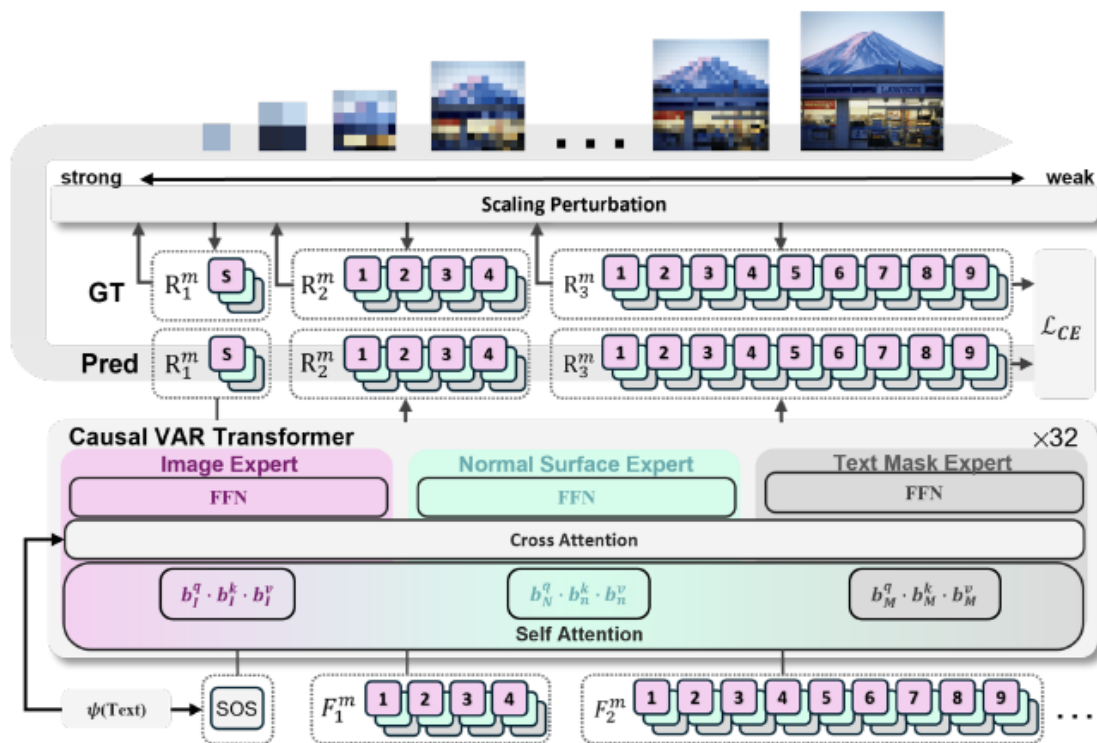
Li-Syun Hsiung, Jun-Kai Tu, Gee-Sern Jison Hsu (IJCAI), 2026 **under review**



PAPER

MART: Multi-Modal AutoRegressive Transformer for High-Fidelity Scene Text Synthesis

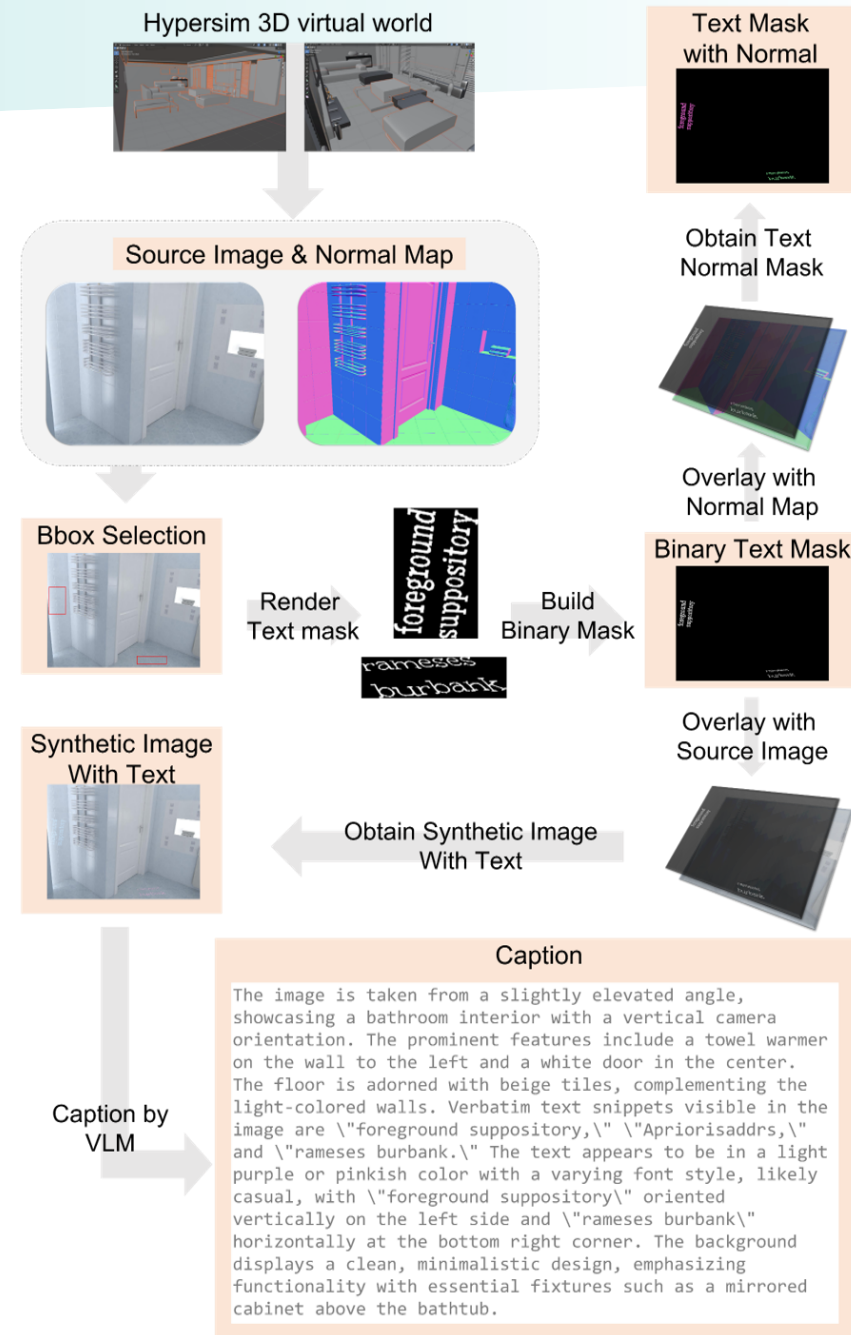
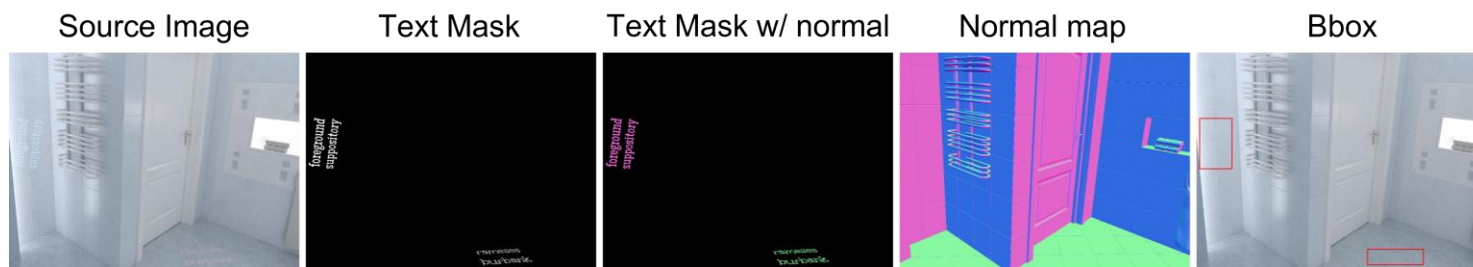
Li-Syun Hsiung, Jun-Kai Tu, Gee-Sern Jison Hsu (IJCAI), 2026 **under review**



PAPER

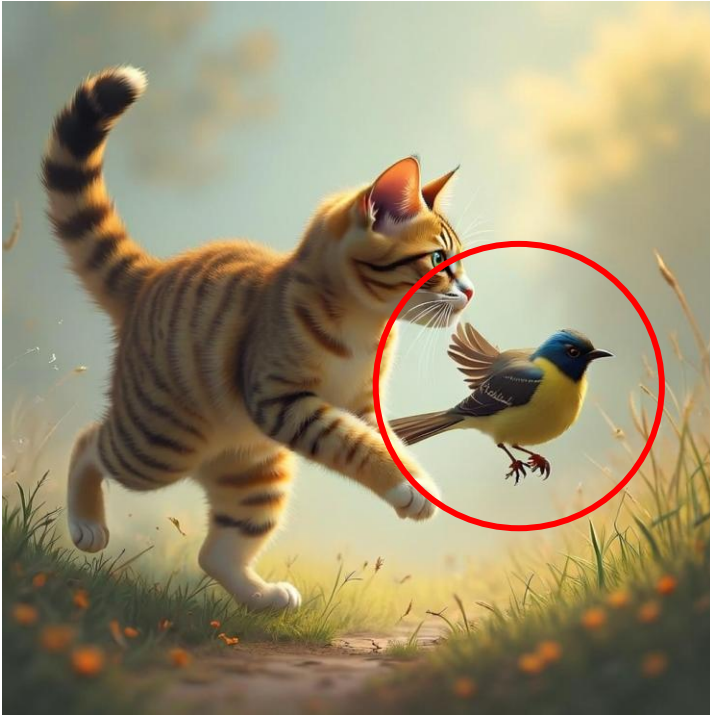
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Ongoing

I_{source}



$I_{naïve}$



I_{target}



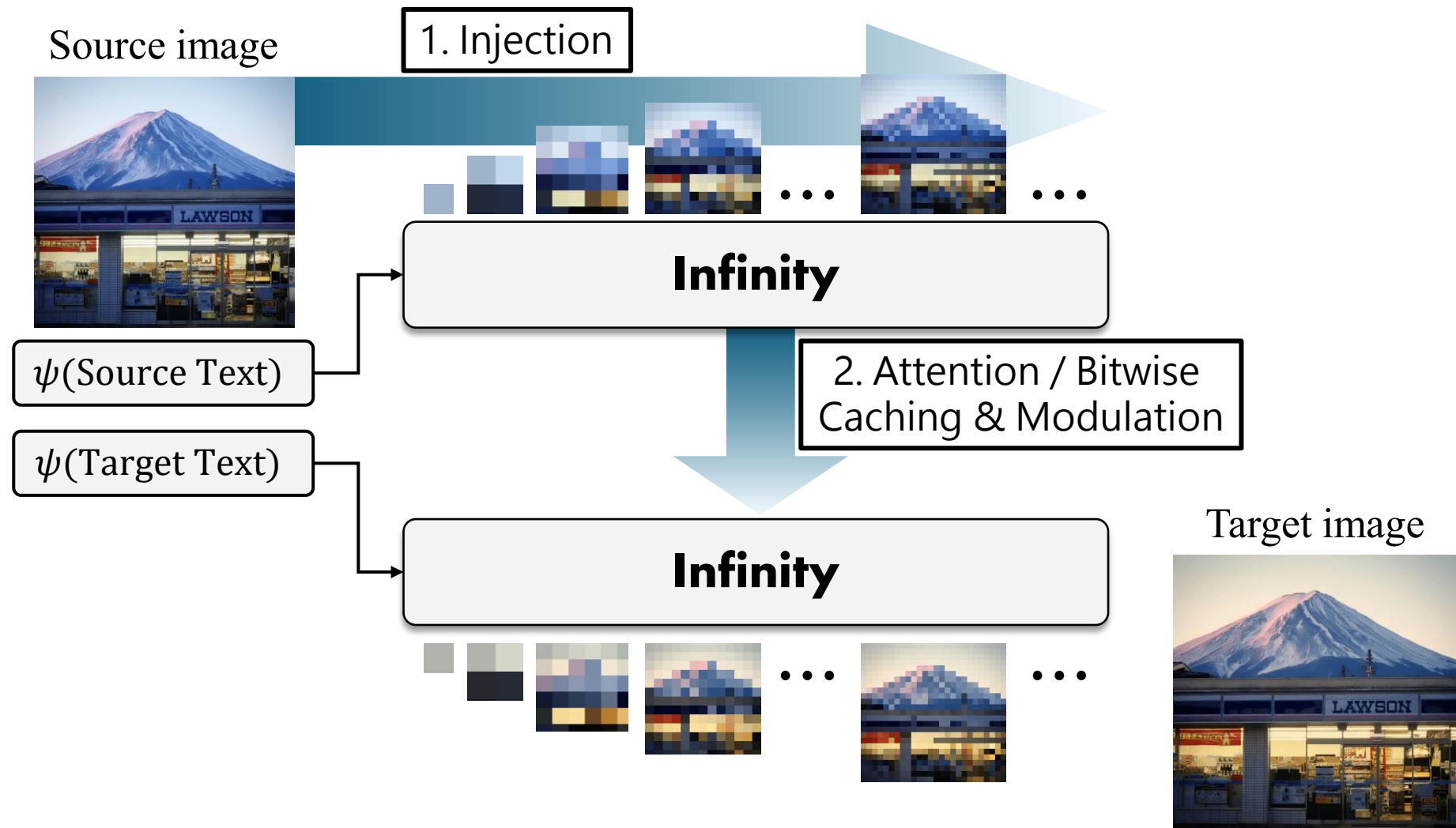
Source prompt:

A cat is chasing a **bird**.

Target prompt:

A cat is chasing a **butterfly**.

Ongoing



Performance



Source prompt: Cute **Shiba Inu** wearing a space helmet.

Target prompt: Cute **cat** wearing a space helmet.



Source prompt: Cute capybara sipping bubble tea in a **cozy cafe**.

Target prompt: Cute capybara sipping bubble tea in a **supermarket**.



Cute baby elephant holding a **sunflower** in a rainy meadow.

Cute baby elephant holding a **lotus** in a rainy meadow.

Performance



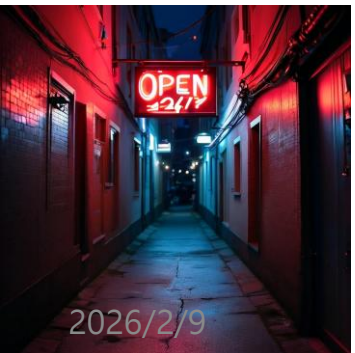
Cute corgi riding a tiny **skateboard** in a sunny park.

Cute corgi riding a tiny **bike** in a sunny park.



Cute penguin wearing a knitted **scarf** on an icy beach.

Cute penguin wearing a knitted **hat** on an icy beach.



A neon-lit alley at midnight with a glowing sign that reads "**OPEN 24/7**".

A neon-lit alley at midnight with a glowing sign that reads "**Always OPEN**".

Performance



Source prompt: A snowy mountain cabin at night with a glowing window sign that reads "**STAY WARM**".

Target prompt: A snowy mountain cabin at night with a glowing window sign that reads "**LEAVE NOW**".

Models I'll Focus On

METAs

 dinov3

 sam3

OCR

 MonkeyOCR

 PaddleOCR

Normal / Depth Estimator

 Lotus

 Marigold

 Depth-Anything-3

T2I models

 VAR

 Infinity

 diffusers

VLMs

 InternVL

 Qwen

Detection

 ultralytics

 detr